Assignment 01 — Walking while carrying a tray with TSID

Optimization and Learning for Robot Control, University of Trento
Daniele Trisotto (267254), Michele Macellari (ID number missing)

All results are produced by `python3 experiments.py qN` (no viewer, default `romeo_conf.py` plus the listed overrides); detailed numbers are in `risultati.md`. $mg = 397.6\,\text{N}$, $t = 0$ is the start of the walk.

1 Walking with double support

Weight transfer. In single support the stance foot carries the whole weight ($\approx mg$, $\pm 4\%$ due to the vertical acceleration of the CoM, $f_N = m(g + \ddot{c}_z)$) and the swing foot 0 N (Fig. 1, Fig. 2). The reference CoP stays under the stance foot ($y = \pm 0.096\,\text{m}$) and, with two contacts, is the average of the feet positions weighted by the normal forces, $p_{CoP} = (f_L p_L + f_R p_R)/(f_L + f_R)$, so $f_L = mg\,(p_{CoP} - p_R)/(p_L - p_R)$: each force is a linear function of the CoP. With double support (DS) the CoP moves smoothly from one foot to the other in 0.2 s and the weight follows gradually (RF $373 \to 35$ N, LF $29 \to 358$ N); without DS the CoP jumps by 19.2 cm in one 2 ms step and the whole weight (≈ 390 N) is transferred in one step (Fig. 3). With a single contact the CoP must stay inside that sole, so without a phase with two contacts it can only jump. Hence the max normal force rate is 11 times larger without DS (195 kN/s vs 17.8 kN/s) and so is the joint torque rate (28.8 vs 6.3 kNm/s). With DS the residual rate comes from two small jumps left by `transition_time = 0` ($\approx$29 N at touchdown, $\approx$35 N at lift-off).

Real robot. Without DS the walk works only because the simulation is ideal: the state is integrated from the QP accelerations (the robot does exactly what TSID computes), contacts are rigid and instantaneous and the touchdown time is exact. On a real robot torques cannot change in 2 ms (actuator bandwidth), the swing foot would take the whole weight during the impact, an early/late touchdown (model errors, uneven ground) would transfer the weight to a foot not yet on the ground, and noisy, delayed force sensors cannot confirm the contact so quickly.

Fall with `transition_time = 0.1` and fix. `removeRigidContact(name, 0.1)` keeps the contact for 0.1 s while decreasing linearly its maximum normal force f_{max} to 0, then removes it; the ramp leaves $f_{max} = 0$ inside the `Contact6d`, and `addRigidContact` re-adds the same object without restoring it. In Fig. 4 the left foot is removed at the start ($f^{LF}_{max} \to 0$ at 1.4 s); at touchdown (2.5 s) the constraint $0 \le f_N \le f_{max} = 0$ forbids it to carry any load, so when the right foot is lifted the robot falls (2.81 s). Fix: `contactRF/LF.setMaxNormalForce(conf.fMax)` in `add_contact_RF/LF` before `addRigidContact`; the walk is then completed (Fig. 5). **The ramp makes little difference** with this gait: same CoM error (2.9/6.8 mm RMS/max), force rate 17.8 $\to$ 15.0 kN/s, torque rate 6.3 $\to$ 4.6 kNm/s. The weight is already transferred by the DS through the CoP: when the ramp starts (2.6 s) the actual force is already below it, so it is active only in the last $\approx$20 ms, removing the 35 N jump at lift-off (Fig. 6); the largest jump left ($\approx$30 N) is at touchdown, which the ramp does not affect. (Without DS the ramp makes the robot fall even with the fix: the only stance foot would be unloaded.)

2 Tray along a sinusoid

Hand task: `TaskSE3Equality`, $K_p = 10$, $K_d = 2\sqrt{K_p}$, mask and frame from the conf, added only if `w_hand>0` or level 0. Reference: $p = c + A\sin\omega t$, $\dot{p} = A\omega\cos\omega t$, $\ddot{p} = -A\omega^2\sin\omega t$.

	0.5 Hz, FF	0.5 Hz, no FF	2 Hz, FF	2 Hz, no FF
Amplitude ratio / phase lag	0.87 / 8.1°	0.44 / 97.2°	0.43 / 7.7°	0.05 / 156.5°
Max CoM displacement	7.3 mm	3.6 mm	54.7 mm	0.9 mm

Without feedforward $\dot{p}^{ref} = \ddot{p}^{ref} = 0$ and only $\ddot{p}^d = K_p(p^{ref} - p) - K_d\dot{p}$ is left: the hand moves only because of the position error, so it lags behind and moves less. It is a critically damped low-pass filter $P/P^{ref} = K_p/(s^2 + K_d s + K_p)$ with $\omega_n = \sqrt{K_p} \approx 0.5$ Hz: at 0.5 Hz ($= \omega_n$) theory gives 0.50/89.6°, at 2 Hz ($\approx 4\omega_n$, attenuation $\propto K_p/\omega^2$) 0.06/151.8°; a frequency sweep matches the theory (Fig. 8, Fig. 7). **With feedforward** ideally $\ddot{e} + K_d\dot{e} + K_p e = 0$ (perfect tracking): the lag is indeed small ($\le 10°$) but the amplitude drops at high frequency. The reason is the CoM: the base is floating, so accelerating the arm and tray pushes the rest of the body (momentum conservation) and moves the CoM, while the CoM task, with the same weight ($w_{com} = w_{hand} = 1$), wants to keep it still; the QP compromises. The required acceleration $A\omega^2$ is 16 times larger at 2 Hz (7.9 vs 0.49 m/s^2): the CoM moves by 5.5 cm (vs 7 mm) and drifts, the hand oscillates around a lower height, and the right ankle reaches its torque limit (Fig. 9, Fig. 7).

3 Walking with the tray

Hand reference: initial height and orientation, x-y = initial hand position + displacement of the CoM reference, velocity/acceleration = those of the CoM reference (zero along z). With `w_hand`=1 instead of 0: max tilt 12.8° $\to$ 0.04°, hand error 66 $\to$ 9 mm, height range 98 $\to$ 14 mm, but CoM error 2.9/6.8 $\to$ 7.1/14.4 mm (RMS/max), mostly along y in single support (Fig. 11, Fig. 10). The cause is not the mass of the arm but the kinematic coupling: the hand pose depends on legs, pelvis and torso, the same DOFs used by the CoM task. Walking makes the body sway ($\pm$8 cm laterally) and, without the hand task, this sway reaches the tray (12.8°, 10 cm); keeping the hand still

requires counteracting the upper-body motion, which also moves the CoM. With equal weights the QP sacrifices part of the CoM tracking, and the 6D hand task leaves less redundancy to the CoM task.

4 Mask and frame of the tray task

Mask $[x, y, z, r_x, r_y, r_z]$, frame	tilt	height range	CoM RMS
full, local = full, world	0.04°	13.5 mm	7.1 mm
[1, 1, 1, 0, 0, 0] position only, local	21.9°	14.1 mm	5.3 mm
[0, 0, 0, 1, 1, 1] orientation only, local	0.02°	85.5 mm	3.1 mm
[0, 0, 1, 1, 1, 0] waiter, local	8.9°	78.4 mm	3.9 mm
[0, 0, 1, 1, 1, 0] waiter, world	0.03°	10.5 mm	3.6 mm

Position only: the orientation is free and the tray tilts up to 22°. Orientation only: the tray is horizontal but the hand moves by 86 mm with the body. The waiter needs constant height and no rotation about the horizontal axes, while x, y and the yaw are free: mask $[0, 0, 1, 1, 1, 0]$, a requirement defined in the *world* frame (Fig. 12, Fig. 13). In the world-aligned frame the mask selects exactly height and tilt: the requirement is met as with the full mask (0.03°, 10.5 mm) with half the CoM error (3.6 vs 7.1 mm), since only 3 DOFs are constrained. In the local frame the same mask selects components along the hand axes, which are not aligned with the world (the hand z axis is 53° from the vertical) and rotate with the hand: freeing r_z frees part of the tilt and freeing x-y frees part of the vertical motion (8.9°, 78 mm). With the full mask the frame is irrelevant (identical results): the error is the same vector in different axes and, with equal gains, its norm does not change.

5 Heavy trays

5 kg tray (Fig. 14): the robot shifts its pelvis by ≈3.5 cm to the left and 2.3 cm backwards for the whole walk (leaning away from the tray, which is in front-right of the pelvis), while the total CoM (robot + tray) is unchanged within a few mm. The payload is in the model, so the CoM task controls the CoM of robot + tray with the same reference; to compensate the 5 kg at ≈0.31 m to the right, the CoM of the robot alone moves by 2.7 cm left and 1.8 cm back, mainly by moving the pelvis. The hand task ($w_{hand} = 100$) keeps the tray in place, so the posture task, with the lowest weight, gives way.

2 kg tray, nominal limits (Fig. 15, Fig. 16): with the hand task as a hard constraint the QP becomes infeasible at 2.23 s, because it conflicts with the **torque bounds**, in particular of `RWristPitch`, the joint carrying the tray (limit $0.6 \times 1.45 = 0.87$ Nm), already at 90% when standing and at 100% just before the failure (right knee at 91%). With $w_{hand} = 100$ the walk is completed: the wrist stays at its limit for 1.24 s and the QP sacrifices hand tracking instead (hand dropping by 13 mm at each step, error up to 32 mm vs 0.4 mm with the empty tray, tilt 0.18°) and slightly the CoM (8.5 vs 7.6 mm RMS). **Hard constraints** should be used for what is physically mandatory and always satisfiable (dynamics, contacts, torque/joint limits): if two of them become incompatible the QP has no solution and the robot falls. Objectives that may conflict with the physics (hand pose, CoM, posture) should be weighted costs: a large weight gives almost strict priority when feasible but lets the task degrade gracefully when it is not.

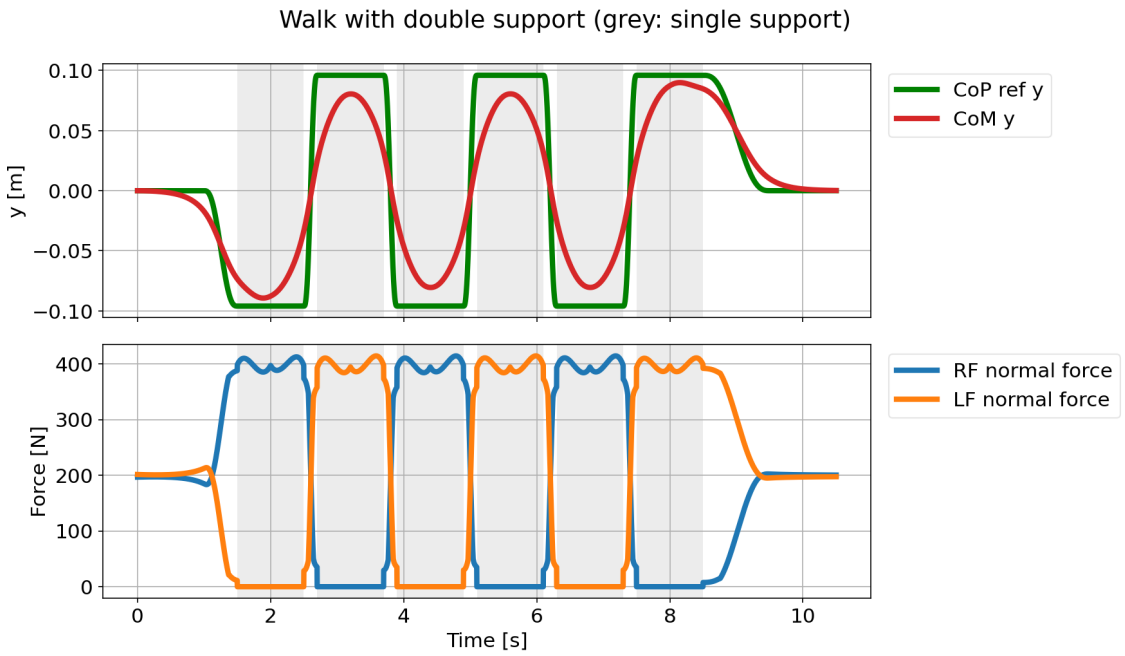


Figure 1: Q1, walk with double support: reference CoP and CoM (lateral), normal forces of the feet.

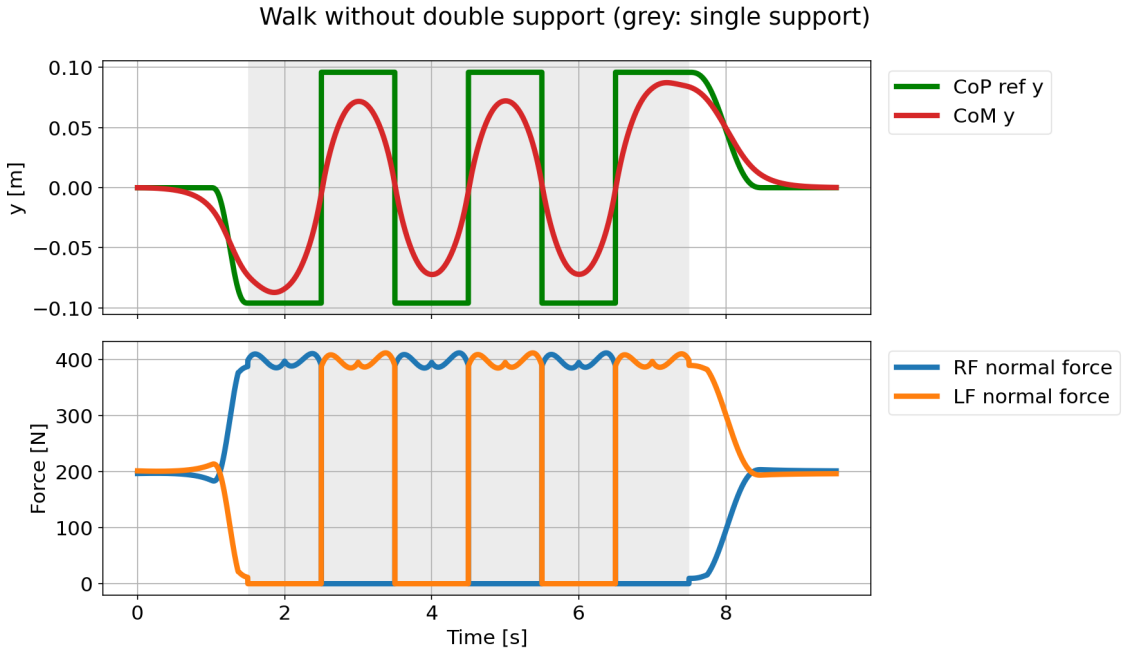


Figure 2: Q1, walk without double support.

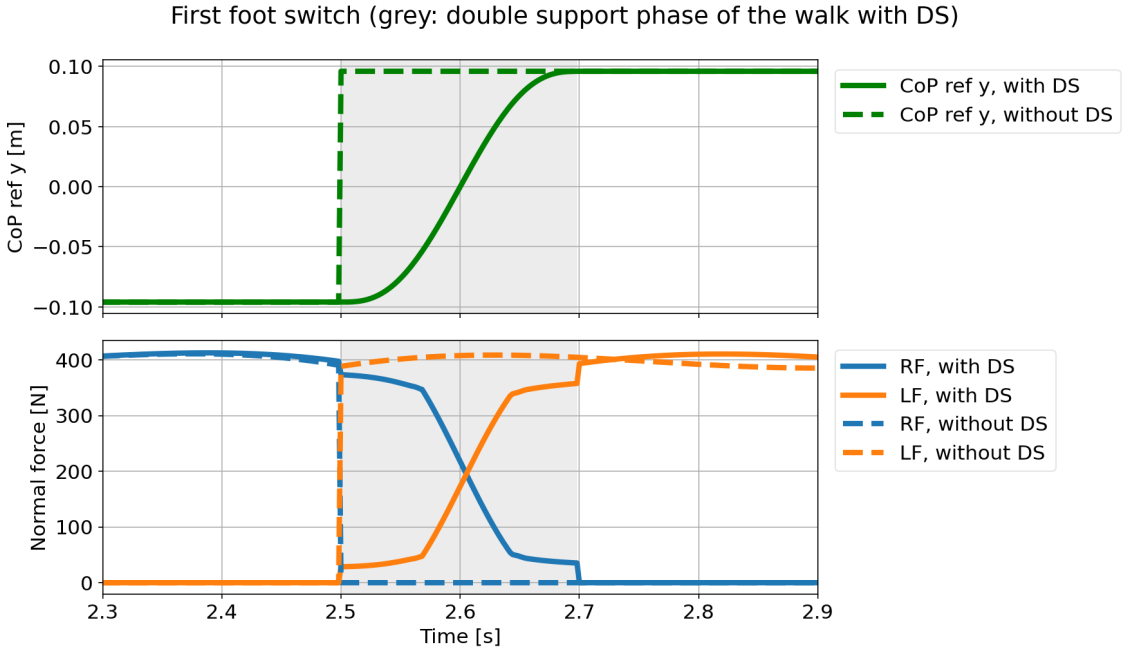


Figure 3: Q1, first foot switch: reference CoP and normal forces with (solid) and without (dashed) double support.

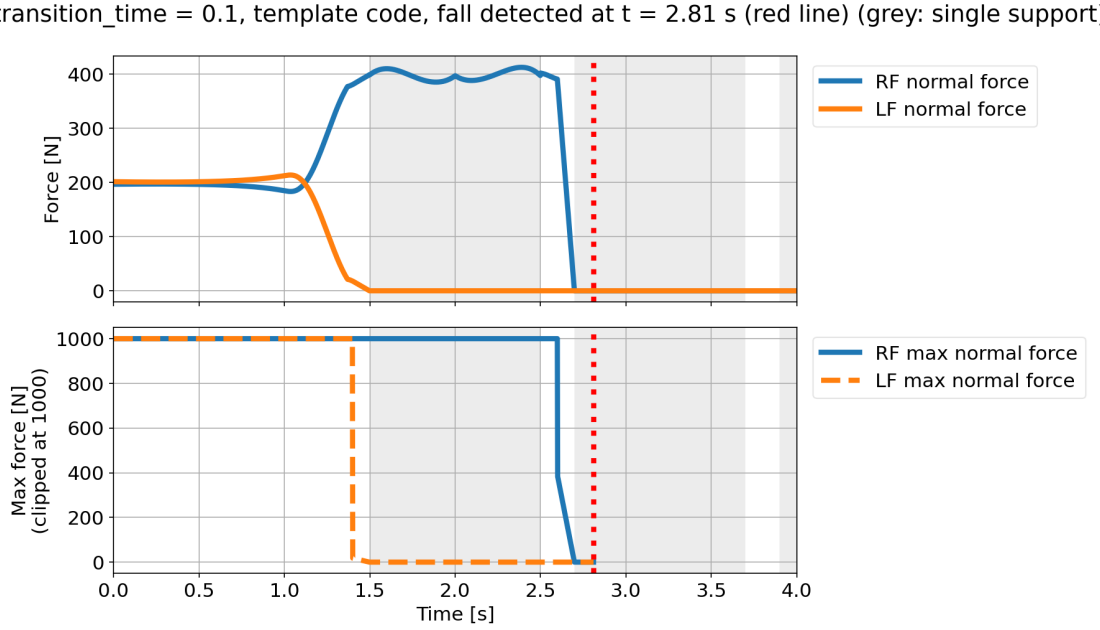


Figure 4: Q1, `transition_time = 0.1` with the template code: normal forces and maximum normal forces until the fall.

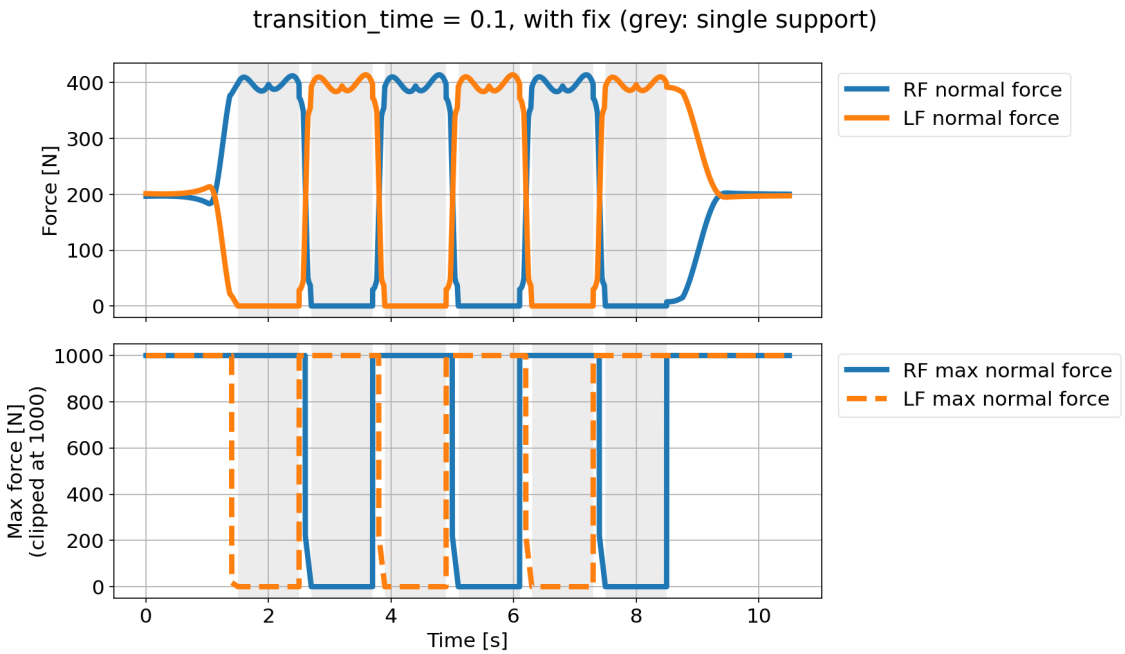


Figure 5: Q1, `transition_time = 0.1` with the fix: f_{max} is ramped to 0 before each lift-off and restored at touchdown.

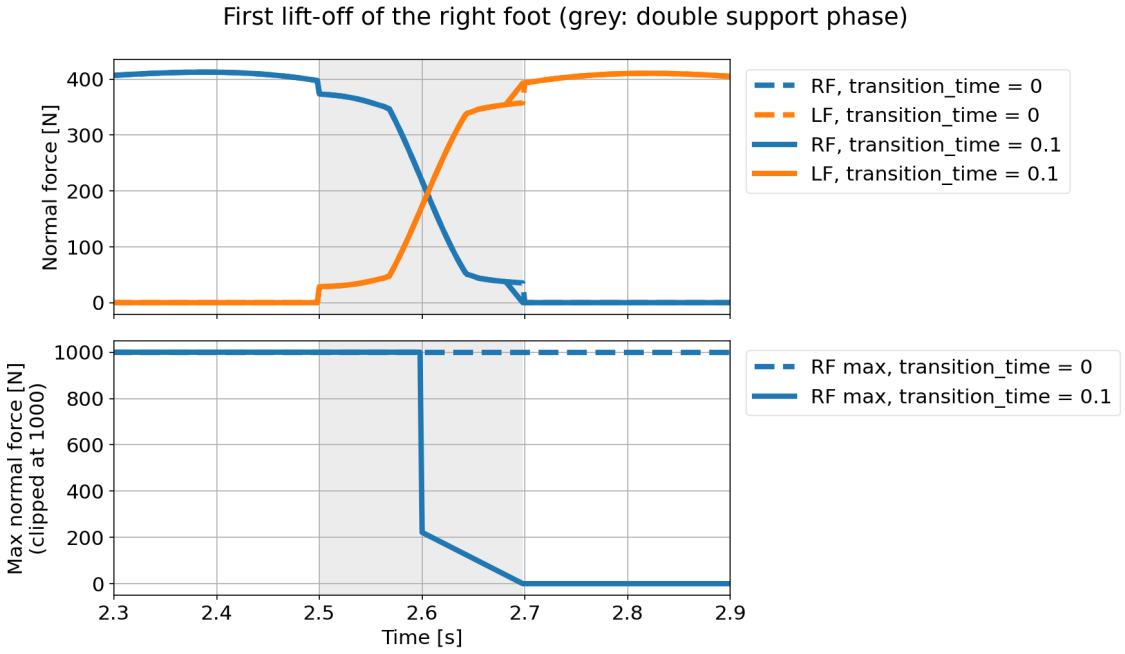


Figure 6: Q1, first lift-off of the right foot with `transition_time` = 0 (dashed) and 0.1 (solid).

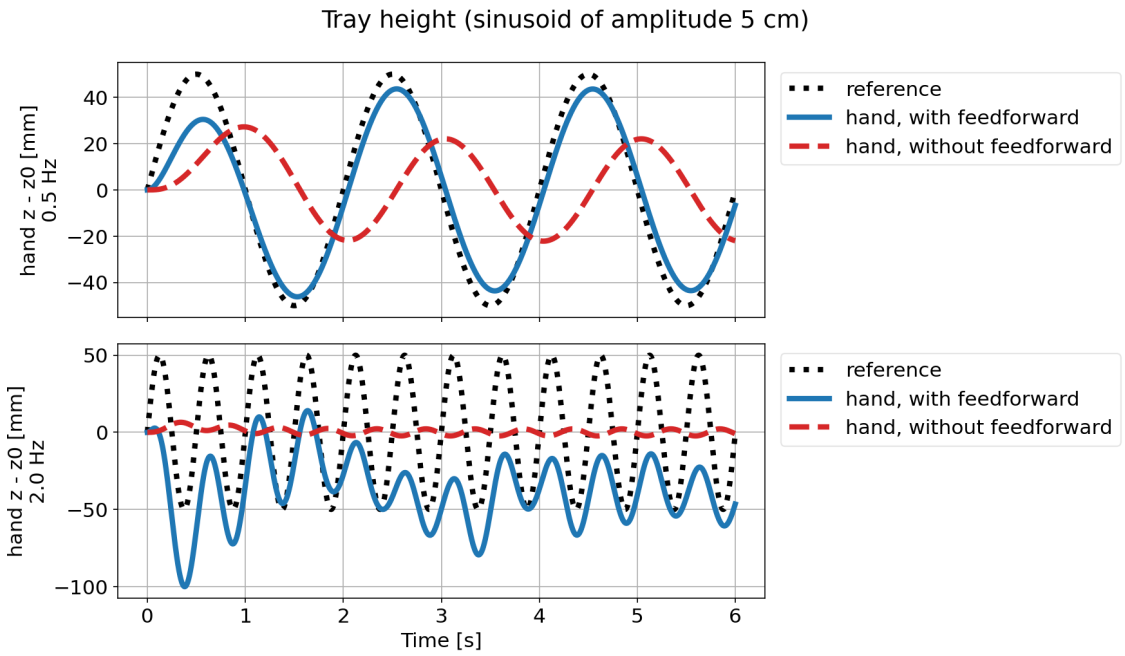


Figure 7: Q2, tray height at 0.5 Hz and 2 Hz, with and without feedforward.

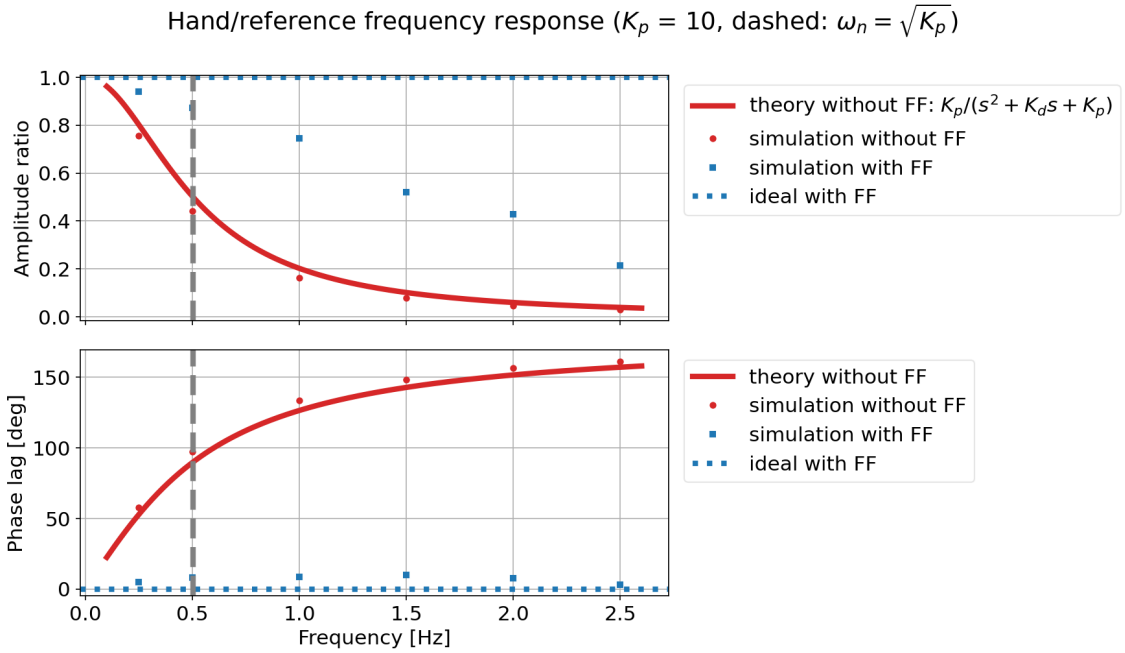


Figure 8: Q2, amplitude ratio and phase lag vs frequency: simulation and PD-only theory.

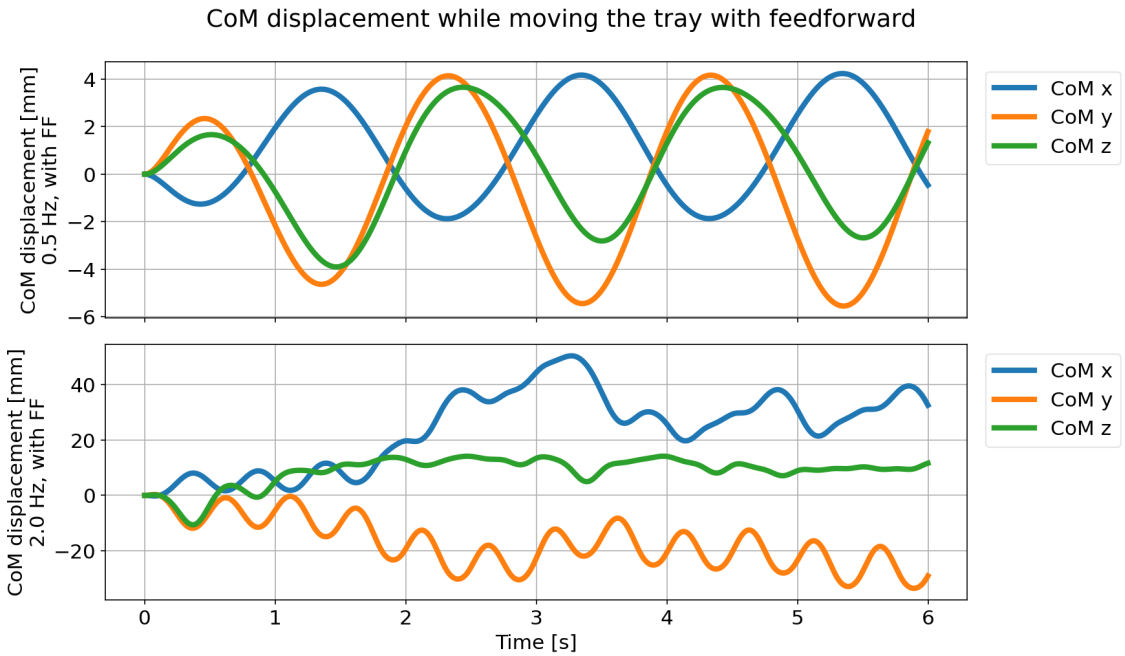


Figure 9: Q2, CoM displacement with feedforward at 0.5 Hz and 2 Hz.

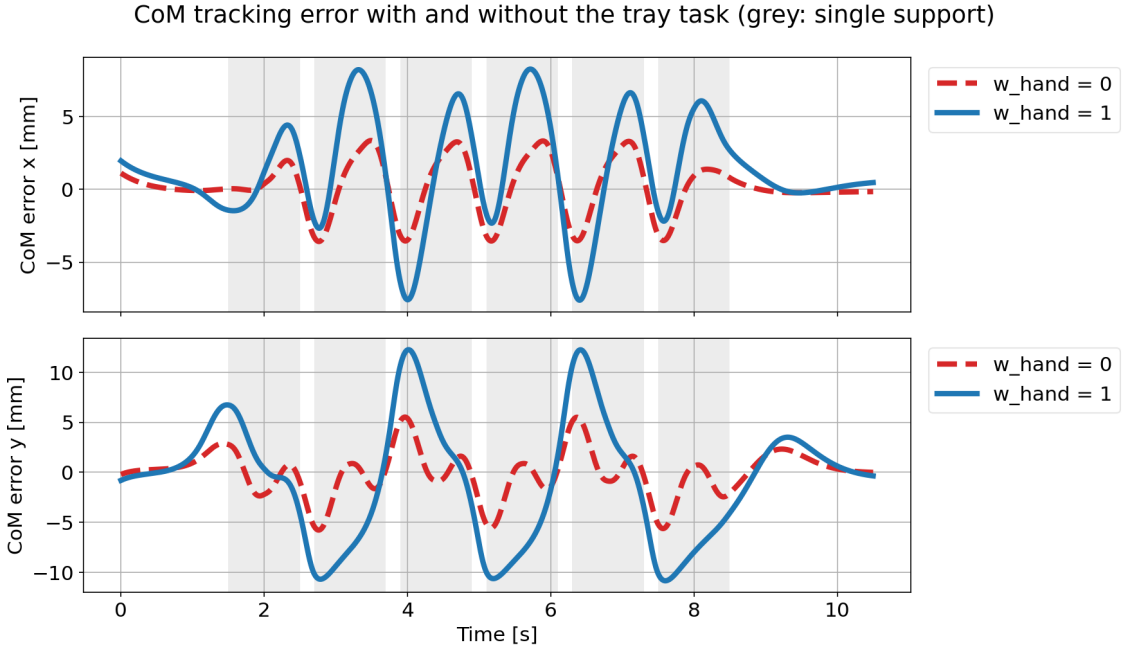


Figure 10: Q3, CoM tracking error with and without the tray task.

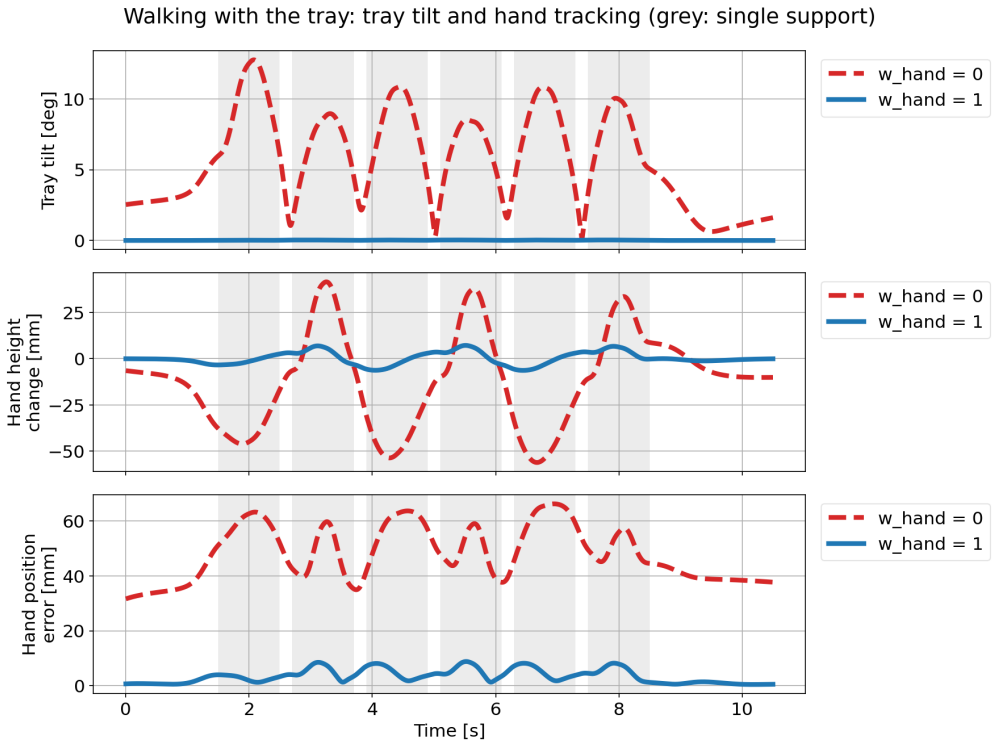


Figure 11: Q3, tray tilt, hand height and hand position error with and without the tray task.

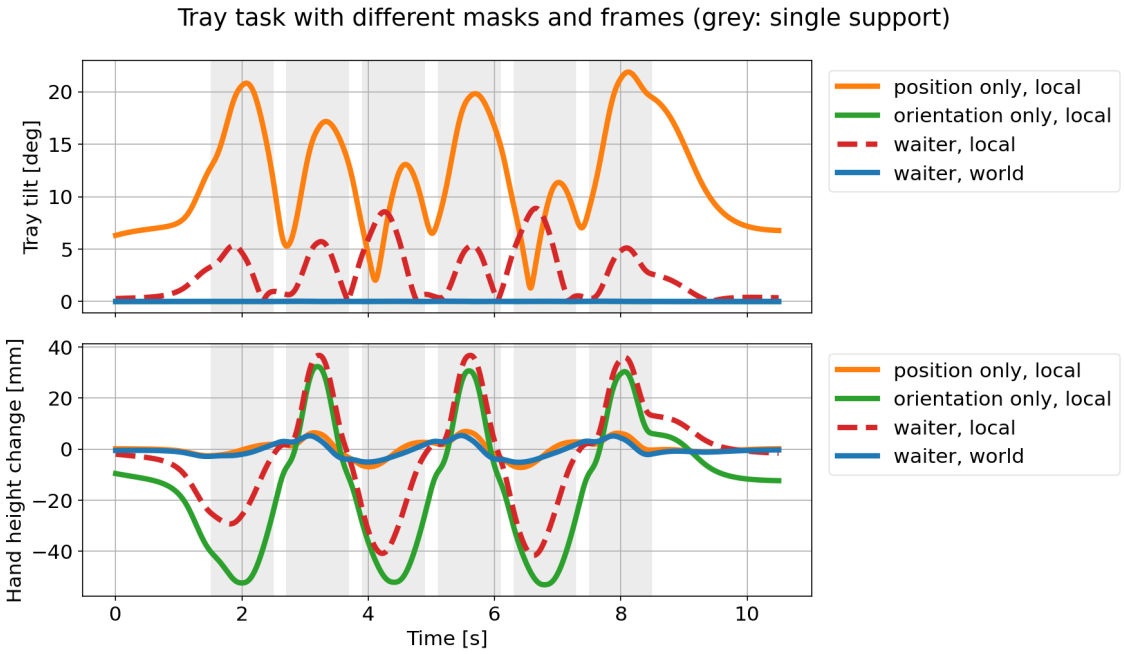


Figure 12: Q4, tray tilt and hand height with different masks and frames.

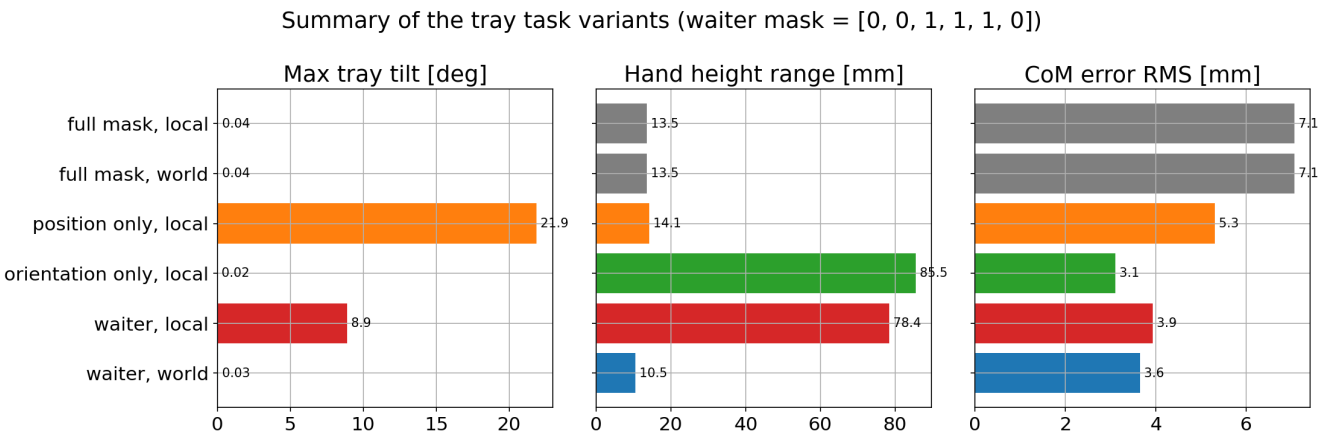


Figure 13: Q4, summary of tilt, height range and CoM error for all masks and frames.

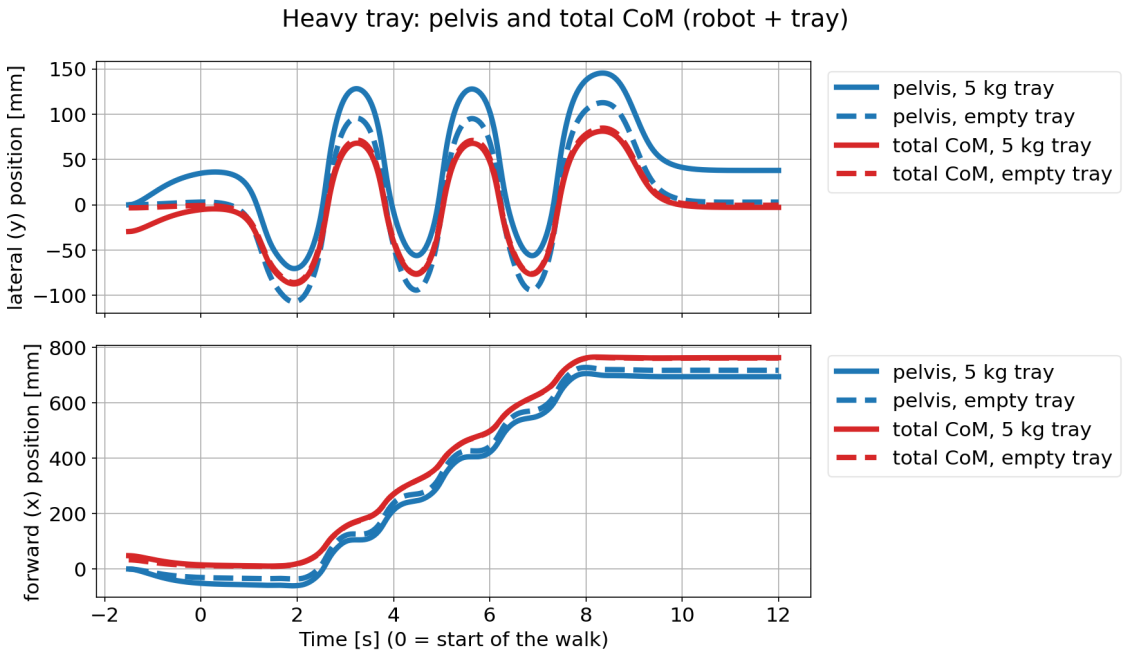


Figure 14: Q5, pelvis and total CoM with the 5 kg tray (solid) and the empty tray (dashed).

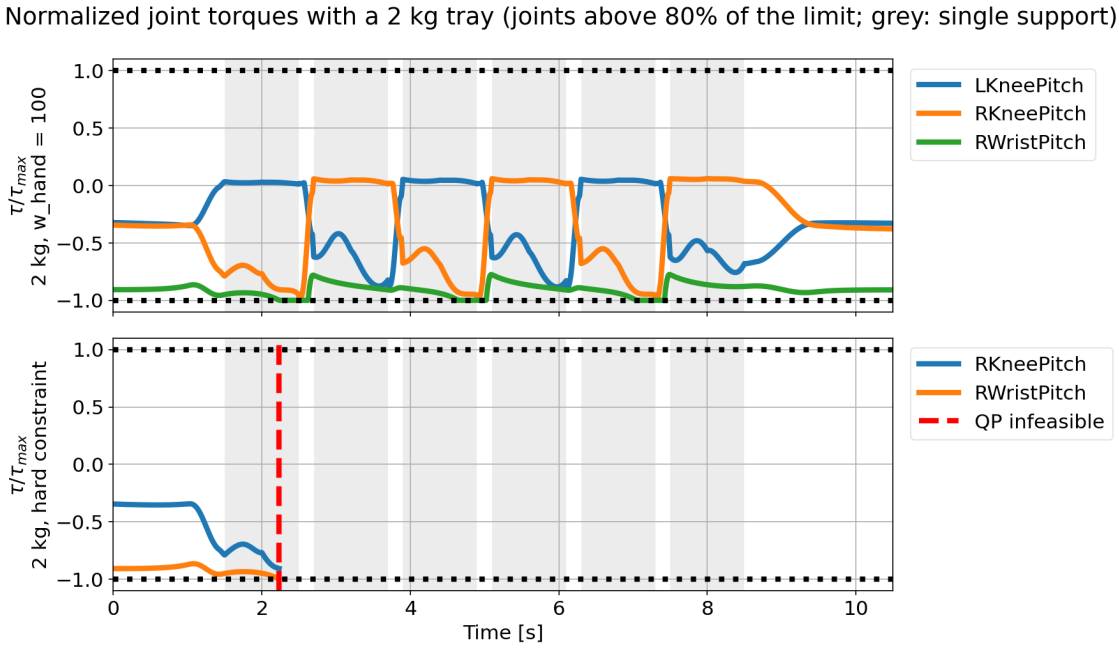


Figure 15: Q5, normalized joint torques with a 2 kg tray: weighted hand task ($w_{hand} = 100$) and hard constraint.

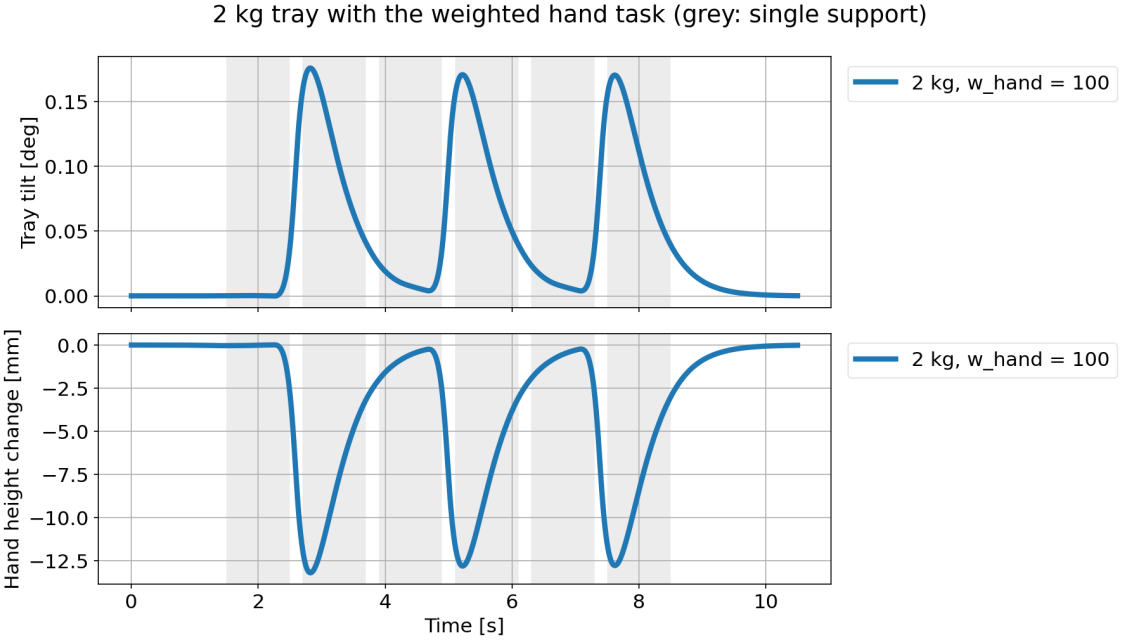


Figure 16: Q5, tray tilt and hand height with a 2 kg tray and $w_{hand} = 100$.